

Chapter 12

Linear PDE with constant coefficient
homogeneous

线性齐次常系数偏微分方程

$u(x, y)$

$$u_{xx} - 2u_{xy} - 3u_{yy} = 0$$

$$u'_{xx} - 2u'_{xy} - 3u'_{yy} = 0 \quad \begin{matrix} u_1(x, y) \\ u_2(x, y) \end{matrix}$$

u_1, u_2 is also the solution

homogeneous: the order of derivative is fixed

$$u_{xxx} + 5u_{xyy} + 2u_{xxy} - u_{yyx} = 0$$

$$D_x u = u_x \quad \text{differential operator}$$

$$D_x^2 u = u_{xx}$$

$$D_x^3 u = u_{xxx}$$

$$D_y u = u_y \quad D_y^2 = u_{yy}$$

$$D_x D_y u = u_{xy}$$

$$u_{xx} - 2u_{xy} - 3u_{yy} = 0$$

$$D_x^2 u - 2D_x D_y u - 3D_y^2 u = 0$$

$$(D_x^2 - 2D_x D_y - 3D_y^2) u = 0$$

$$(D_x - 3D_y)(D_x + D_y) u = 0$$

$$\left[\begin{array}{l} (D_x + D_y) u = u_x + u_y \\ D_x \cdot (D_x + D_y) u - 3D_y (D_x + D_y) u \end{array} \right] u$$

$$= u_{xx} + u_{xy} - 3(u_{xy} + u_{yy})$$

$$= u_{xx} - 2u_{xy} - 3u_{yy}$$

$$(D_x - 3D_y)(D_x + D_y) u = 0$$

$$A - B = 0$$

$$A = 0, B = 0$$

$$\left\{ \begin{array}{l} (D_x - 3D_y) u = 0 \\ (D_x + D_y) u = 0 \end{array} \right.$$

$$u_x + u_y = 0 \Rightarrow \phi(y-x) \quad \phi'(y-x) \cdot (-1) + \phi'(y-x) \cdot 1 = 0$$

$$u_x - 3u_y = 0 \Rightarrow \psi(3x+y) \quad \phi'(3x+y) \cdot 3 - 3\phi'(3x+y) \cdot 1 = 0$$

$$u_x - \alpha_1 u_y = 0 \quad \phi(\alpha_1 x + y) = 0$$

$$u_{xx} - 5 u_{xy} - 6 u_{yy} = 0$$

$$(D_x^2 - 5 D_x D_y - 6 D_y^2) u = 0$$

$$= (D_x - 6 D_y)(D_x + D_y) u = 0$$

$$(D_x - 6 D_y) u = 0 \quad \phi(6x + y)$$

$$(D_x + D_y) u = 0 \quad \psi(x-y) \quad \phi'(6x+y) \cdot 6 - 6 \phi'(6x+y)$$

$$u(x-y) = \phi(6x+y) + \psi(x-y)$$

$$u_{xx} - 2 u_{xy} + 2 u_{yy} = 0$$

$$(u_x - \alpha_1 u_y)(u_x - \alpha_2 u_y) = 0 \quad \alpha_1, \alpha_2$$

$$\alpha_1^2 - 2\alpha_1 + 2 = 0$$

$$u_{xx} - u_{xy} = 0$$

$$\left\{ \begin{array}{l} (D_x^2 - D_x D_y) u = 0 \\ D_x (D_x - D_y) u = 0 \\ D_x u = 0 \\ (D_x - D_y) u = 0 \end{array} \right. \quad \phi$$

$$(D_x - D_y) u = 0$$

$$(A D_x - B D_y) u = 0 \quad \phi(Bx + Ay)$$

$$D_x u = 0 \quad A=1 \quad B=0 \quad \phi(y)$$

$$u_{xx} - 3 u_{xy} + 2 u_{yy} = 0$$

PDE

$$\left\{ \begin{array}{l} u_{xx} - 3 u_{xy} + 2 u_{yy} = 0 \quad f(x) \\ e^x e^{2x} \quad \lambda^2 - 3\lambda + 2 = 0 \quad 1, 2 \quad \text{ODE} \end{array} \right.$$

$$u_p + c_1 e^x + c_2 e^{2x}$$

$$\frac{c_1 e^x + c_2 e^{x \cdot x}}{c e^x}$$

$$f(x): \quad e^{ix} \quad \leftarrow \quad \frac{e^{ix} - e^{-ix}}{2}$$

$$p(x) = 2x^2 + 3$$

Non-zero input

$$(1) f(x, y) = e^{ax+by}$$

$$u_{xx} - 3u_{xy} + 2u_{yy} = e^{ax+by}$$

$$u = A e^{ax+by} \quad \begin{matrix} \text{--- } \mathcal{L}(D_x D_y) \\ = D_x^2 - 3D_x D_y + 2D_y^2 \end{matrix}$$

$$u_{xx} = A \cdot a^2 e^{ax+by}$$

$$u_{xy} = A \cdot a b e^{ax+by}$$

$$u_{yy} = A \cdot b^2 e^{ax+by}$$

$$(A \cdot a^2 - 3Aab + 2Ab^2) e^{ax+by} = e^{ax+by}$$

$$A(a^2 - 3ab + 2b^2) = 1$$

$$\mathcal{L}(a \cdot b)''$$

$$A = \frac{1}{\mathcal{L}(a \cdot b)}$$

$$\mathcal{L}(a \cdot b) \neq 0$$

$$f(x, y) = e^{ax+by}$$

$$u_p(x, y) = \frac{1}{\mathcal{L}(a \cdot b)} e^{ax+by}$$

$$e^{5x+7y}$$

$$\mathcal{L}(a \cdot b) = a^2 - 3ab + 2b^2 = 5^2 - 3 \cdot 5 \cdot 7 + 2 \cdot 7^2$$
$$= 25 - 105 + 98$$

$$u_p(x, y) = \frac{1}{18} e^{5x+7y} = 18$$

$$e^{ax+b} = f(x, y)$$

$$A u_x - B u_y = 0$$

$$\mathcal{L}(a \cdot b) = 0$$

$$A \cdot a - B \cdot b = 0$$

$$b D_x - a D_y = 0$$

$$b \cdot a - a \cdot b = 0$$

$$(b D_x - a D_y) \cdot u = e^{ax+by}$$

$$\mathcal{L}(a \cdot b) = 0$$

$\frac{x}{b} e^{ax+by}$	$\frac{y}{a} e^{ax+by}$
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Ex 12.4

$$(2D_x - 3D_y)(D_x + D_y)u = 5e^{x-y}$$

$$(D_x + D_y) \underbrace{(2D_x - 3D_y)u}_v = 5e^{x-y}$$

$$\begin{cases} (D_x + D_y)v = 5e^{x-y} \\ (2D_x - 3D_y)u = v \end{cases}$$

$$\begin{aligned} (D_x + D_y)v &= 5e^{x+y} \\ \mathcal{L}(a \cdot b) &= 5e^{ax+by} \quad a=1, b=1 \\ \therefore a+b &= 2 \\ v_p(x,y) &= \frac{5}{2} e^{x+y} \end{aligned}$$

$$\begin{cases} (2D_x - 3D_y)u = \frac{5}{2} e^{x+y} \\ \mathcal{L}(a \cdot b) = 2a - 3b = 2 - 3 = -1 \end{cases} \quad a=1, b=1$$

$$u_p = \frac{1}{-1} \frac{5}{2} e^{x+y}$$

$$e^{ax+by}$$

$$a \cdot b$$

$$f(x,y) = 12xy$$

algebra

Ex 12.5

$$(D_x^2 - 2D_x D_y + D_y^2)u = 12xy$$

$$(D_x - D_y)^2 u = 12xy$$

$$u = \frac{1}{(D_x - D_y)^2} 12xy$$

$$= \frac{1}{D_x^2} \cdot \frac{1}{\left(1 - \frac{D_y}{D_x}\right)^2} 12xy$$

$$= \frac{1}{D_x^2} \cdot \left(1 - \frac{D_y}{D_x}\right)^{-2} 12xy \quad \leftarrow 12xy + 12x^2$$

$$(1-x)^{-2} = 1 + 2x + C_3 x^2 + \dots$$

Taylor expansion at 0

$$(-2)(1-x)^{-3} \Big|_{x=0} = (-2)(1) = -2$$

$$\left(1 - \frac{D_y}{D_x}\right)^{-2} = 1 + 2 \frac{D_y}{D_x} + C_3 \frac{D_y^2}{D_x^2} + \dots$$

$$\left(1 - \frac{D_y}{D_x}\right)^{-2} (12xy) = 12xy + 2\left(\frac{1}{D_x} \cdot D_y\right) 12xy + C_3 \frac{1}{D_x^2} D_y^2 (12xy) + D_y^3$$

$$D_y (12xy) = 12x$$

$$\frac{1}{D_x} 12x = \underline{6x^2}$$

$$12x = D_x (6x^2)$$

$$\rightarrow = (12xy + 12x^2)$$

$$\frac{1}{D_x^2} (12xy + 12x^2) = 2x^3y + \underline{x^4}$$

$$\frac{1}{D_x^2} g = f$$

$$g = D_x^2 f$$

$$\frac{1}{D_x^2} \cdot 0 = f$$

$$u_{xx} - u_{yy} = xy - x \quad \leftarrow \frac{1}{6} x^3 (y-1)$$

$$u_{xx} - u_{yy} = xy - x \quad \in \frac{1}{6} x'(y-1)$$

$$(D_x^2 - D_y^2) u = xy - x$$

$$u_{xx} = x(y+1)$$

$$u_{yy} = 0$$

$$u = \frac{1}{D_x^2 - D_y^2} (xy - x)$$

$$\textcircled{1} \quad = \frac{1}{D_x - D_y} \frac{1}{D_x + D_y} (xy - x)$$

$$\frac{1}{D_x + D_y} (xy - x) = f(x, y)$$

$$xy - x = (D_x + D_y) f(x, y)$$

$$= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$$

$$u = \frac{1}{D_x^2 - D_y^2} (xy - x)$$

$$= \frac{1}{D_x^2} \left(\frac{1}{1 - \frac{D_y^2}{D_x^2}} (xy - x) \right) = (xy - x)$$

$$\frac{1}{1-t} = 1 + t + t^2 + \dots$$

$$\frac{1}{1 - \left(\frac{D_y^2}{D_x^2}\right)} = \left[1 + \frac{D_y^2}{D_x^2} + \left(\frac{D_y^2}{D_x^2}\right)^2 + \dots \right] (xy - x)$$

$$= xy - x$$

$$+ \frac{1}{D_x^2} D_y^2 (xy - x) > 0$$

$$+ \frac{1}{D_x^2} \cancel{D_y^2 (xy-x)}$$

$$+ \dots 0$$

$$\frac{1}{D_x^2} (xy-x) = f$$

$$f = +\frac{1}{6} x^3 (y-1)$$

$$u_{xx} + u_{yy} = x^2 + xy$$

$$(D_x^2 + D_y^2) u$$

$$u = \frac{1}{D_x^2 + D_y^2} x^2 + xy$$

$$= \frac{1}{D_x^2} \frac{1}{1 + \frac{D_y^2}{D_x^2}} (x^2 + xy)$$

$$\left(1 - \frac{D_y^2}{D_x^2} + \dots \right) (x^2 + xy)$$

$$= x^2 + xy - \frac{D_y^2}{D_x^2} (x^2 + xy)$$